

Keaton Kraiger

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Education

- Pennsylvania State University**, PhD in Computer Science Engineering Expected Dec. 2026
- Advisors: Dr. Yanxi Liu and Dr. Robert T. Collins
 - Research topics: Computer vision, multimodal learning, and learning human dynamics from video
- Portland State University**, BS in Computer Science Aug. 2016 – June 2020
- Advisor: Dr. Dan Hammerstrom
 - Graduated Magna Cum Laude

Research Interests

My research focuses on machine learning and its applications to multimodal data, including video, sensor, and motion-capture inputs. I am particularly interested in developing AI models that can understand, predict, and generate complex patterns in dynamic data, with applications spanning computer vision, graphics, and intelligent automation.

Publications

- FootFormer: Estimating Stability from Visual Input** Nov. 2025
Keaton Kraiger, Jingjing Li, Skanda Bharadwaj, Jesse Scott, Robert T. Collins, Yanxi Liu.
BMVC 2025
- Novel 3D Scene Understanding Applications From Recurrence in a Single Image** Aug. 2023
Shimian Zhang, Skanda Bhara, **Keaton Kraiger**, Yashasvi Asthana, Hong Zhang, Robert T. Collins, Yanxi Liu.
arXiv:2210.07991
- Biomechanically Justified Benchmark for Spatiotemporal Quantification of Human Motion Dynamics** Under Review
Keaton Kraiger, Skanda Bharadwaj, Spencer Yao Yi Huang, Jesse Scott, John H Challis, Robert T. Collins, Yanxi Liu.
- Implicit Learning of Explicit Class-Topology** Under Review
Yanxi Liu, Christopher Funk, *Keaton Kraiger*.
- Action Recognition from Multimodal Human Motion Data** In Preparation
Keaton Kraiger, Addison Petro, Yuan Gao, Jeff Koumba, Yanxi Liu, Robert T. Collins.

Presentations and Posters

- Learn From Human Eyes: Zero-Shot Recurring Pattern Detection on a Multi-Perception Benchmark** May 2025
Shimian Zhang, **Keaton Kraiger**, Skanda Bharadwaj, Robert Collins, Yanxi Liu.
Best Oral Presentation Award
Symmetry 2025 Abstract Track | Slides
- Vision to Dynamics** Feb. 2025
Keaton Kraiger, Skanda Bharadwaj, Yanxi Liu, Robert T. Collins.
NYC Vision Day 2025 | Poster
- Vision to Dynamics** April 2024
Keaton Kraiger, Yuan Gao, Jeff Koumba, Yanxi Liu, Robert T. Collins.
NYC Vision Day 2024 | Poster
- Humanoid Robots (Real and Simulated)** Feb. 2024
Yuan Gao, Jeff Koumba, **Keaton Kraiger**.

The Applications of Grid Cells in Computer Vision

April 2019

Keaton Kraiger, Dan Hammerstrom.

Portland State University Student Research Symposium | Poster

Research Experience

Graduate Research Assistant, Pennsylvania State University – State College, PA

Jan. 2021 – Present

Laboratory for Perception, Action, and Cognition (LPAC)

- Advisors: Dr. Yanxi Liu and Dr. Robert Collins
- Architected and deployed a ROS2-based multimodal capture system combining synchronized RGB-D camera arrays, Meta Project Aria smart glasses, and insole pressure sensors
- Built and optimized machine learning models to predict human stability from video, motion capture, and foot pressure data, achieving state-of-the-art results in Python/PyTorch
- Released a curated dataset of 60K+ frames with automated cleaning and synchronization tools for motion capture, video, and pressure modalities
- Created a class-agnostic object detection framework enabling downstream scene understanding in unstructured environments, improving language model generation

Undergraduate Research Assistant, Portland State University – Portland, OR

Dec. 2018 – Jan. 2020

Biologically-Inspired Computing Lab (BICL)

- Advisor: Dr. Dan Hammerstrom
- Implemented and benchmarked biologically-inspired object detection algorithms in Python, achieving position and scale invariance
- Processed image datasets using a grid cell model to improve object detection and image classification
- Evaluated CNN and capsule network performance when integrated with the grid cell model

Undergraduate Research Fellow, Portland State University - Portland, OR

Nov. 2018 – May 2019

- One of four students selected for a year-long Undergraduate Research Fellowship in computer vision
- Participated in training to develop research design, writing, and presentation skills

Work Experience

Head Teaching Assistant, Pennsylvania State University – State College, PA

Aug. 2021 – Dec. 2023

- Head TA for Vision and Language, Computer Vision I & II, and Machine Learning
- Designed and implemented course projects focused on vision, deep learning, and reinforcement learning

Technical Course Specialist, Portland State University – Portland, OR

Sep. 2017 – June 2020

- Led homework recitation sessions on undergraduate programming assignments and algorithm development
- Provided feedback and grades on programming and written assignments

Additional Experience And Awards

Undergraduate Research & Mentoring Program (URMP)

- Selected to participate in the URMP at Portland State, receiving funding to conduct research with a faculty mentor and receive training on conducting research

Outstanding Teaching Assistant Award

- Recognized by the CSE college for Vision and Language Spring 2023 TA

Conference Reviewer

- Reviewed for major computer vision conferences—including ICCV, ECCV, and CVPR—for four consecutive years